Demo 06 — Advanced: Multi-model executive review

Description

Simulates an executive quarterly review spanning two interconnected business models — a revenue forecast and a hiring plan. Demonstrates how all three skills (modeler, analyst, planner) work together across multiple models.

1. Create a revenue forecast model with real-looking SaaS numbers
2. Create a hiring plan model tracking headcount and cost-per-hire
3. Populate both with quarterly data (including a Q3 dip in revenue)
4. Analyze both: summary + anomaly detection
5. Build a revenue recovery plan with milestones
6. Compare hiring scenarios: freeze vs steady vs accelerate
7. Generate strategic recommendations tying both models together
8. Clean up

Demo

> *Create a revenue forecast model called demo-exec-revenue for Q1–Q4 2026 with Revenue and Costs, assuming 15% growth and 55% cost ratio, including actuals for Q1 through Q3: Revenue $1.2M, $1.4M, $0.9M and Costs $660K, $770K, $720K.*

The model demo-exec-revenue has been created with the following setup:

- Dimensions: Q1 2026, Q2 2026, Q3 2026, Q4 2026
- Metrics: Revenue (USD), Costs (USD)
- Assumptions: 15% revenue growth, 55% cost ratio Actuals (entered as provided):
- Q1: Revenue $1,200,000 | Costs $660,000
- Q2: Revenue $1,400,000 | Costs $770,000
- Q3: Revenue $900,000 | Costs $720,000 Forecast (Q4 2026):
- Revenue: $1,035,000 (15% growth over Q3)
- Costs: $569,250 (55% of Q4 Revenue) Let me know if you'd like to see or further modify the model!

> *Show me the demo-exec-revenue overview.*

Here is the overview of the demo-exec-revenue model: Model: demo-exec-revenue Updated: 2026-03-22

	Q1 2026	Q2 2026	Q3 2026	Q4 2026
Revenue (USD)	1,200,000	1,400,000	900,000	1,035,000
Costs (USD)	660,000	770,000	720,000	569,250
Assumptions:				

- revenue_growth_rate: 0.15
- cost_ratio: 0.55 Let me know if you need further details or want to make changes!

> *Analyze the revenue trend for demo-exec-revenue.*

Here is the period-over-period revenue trend for the demo-exec-revenue model: Metric: Revenue (USD)

Period	Before	After	Change
Q1 2026 → Q2 2026	1,200,000	1,400,000	+16.7%
Q2 2026 → Q3 2026	1,400,000	900,000	-35.7%
Q3 2026 → Q4 2026	900,000	1,035,000	+15.0%
Highlights:			

- Revenue grew by 16.7% from Q1 to Q2.
- There was a significant drop (-35.7%) from Q2 to Q3.
- A recovery is forecast for Q4, with 15.0% growth over Q3. If you want to explore the reasons behind these changes or need further analysis, let me know!

> *Detect anomalies in demo-exec-revenue — something happened in Q3.*

No anomalies were detected in the demo-exec-revenue model using a 50% deviation threshold, even in Q3 2026. This suggests that while Q3 revenue dropped significantly, it was not extreme enough versus the mean to be classified as an anomaly by the current criteria. If you want a more sensitive analysis, we can lower the threshold or review the data in a different way. Let me know how you'd like to proceed!

> *Create a hiring plan model called demo-exec-hiring for Q1–Q4 2026, tracking Headcount and CostPerHire, with Q1–Q3 data.*

The model demo-exec-hiring has been created for Q1–Q4 2026 with the following metrics:

- Headcount (FTE)
- CostPerHire (USD) Data entries are ready for Q1 through Q3 2026; Q4 is left blank for future planning. Please provide the actual numbers for Headcount and CostPerHire in Q1–Q3, or let me know if you'd like to proceed with any assumptions or next steps!

> *Run a summary analysis on demo-exec-hiring.*

The summary analysis for demo-exec-hiring did not complete successfully because the model currently does not have any actual data entered for Headcount or CostPerHire in any quarter (all values are blank or zero). To generate a meaningful summary, please provide the actual Headcount and CostPerHire values for Q1, Q2, and Q3 2026. Let me know the numbers, and I will update and re-run the analysis for you!

> *Build a revenue recovery plan for demo-exec-revenue targeting $1.8M by Q4 2026.*

Here's a revenue recovery plan for demo-exec-revenue targeting $1.8M by Q4 2026: Action Plan: demo-exec-revenue Target: Revenue = $1,800,000 by Q4 2026 Current State:

- Latest Revenue (Q4): $1,035,000
- Peak Revenue (Q2): $1,400,000 Gap Analysis:
- Reference (Q3): $900,000
- Target: $1,800,000
- Gap: +$900,000 (+100.0%) Required Growth Rate:
- +100.0% growth per period over 1 period (Q4) Milestone Plan:

Quarter	Revenue (USD)	Status
Q1 2026	$1,200,000	✓ actual
Q2 2026	$1,400,000	✓ actual

Q3 2026	$900,000	✓ actual
Q4 2026	$1,035,000	← current/target
Recommended Actions:		

- Update revenue_growth_rate to reflect the +100% growth required from Q3 to Q4.
- Run a simulation using the new target growth rate to preview new potential milestones.
- After simulation, update the model to commit to the revised recovery plan. Would you like to simulate this revised growth scenario or take other action on the model?

> *Compare three hiring scenarios for demo-exec-hiring in Q4: Freeze at 50, Steady Growth at 55, and Accelerate Hiring at 65.*

The scenario comparison for demo-exec-hiring could not be fully executed because the underlying model does not have actual data for preceding Headcount metrics or CostPerHire, causing a calculation error. This means the comparison table for the Q4 headcount scenarios (Freeze at 50, Steady Growth to 55, Accelerate Hiring to 65) could not be completed. To perform a valid scenario analysis:

- Please provide actual Headcount values for Q1–Q3 2026, and if available, CostPerHire for those periods.
- With this data, the tool will be able to meaningfully compare the Q4 hiring scenarios and outline their impacts. Let me know the missing Q1–Q3 headcount and cost figures, or if you want to proceed based on default/assumed values!

> *Simulate demo-exec-revenue with aggressive 20% growth and a 50% cost ratio, save as demo-exec-revenue-recovery for analysis.*

The simulation with an aggressive 20% revenue growth rate and a 50% cost ratio has been completed and saved as the model demo-exec-revenue-recovery. Here's the before/after comparison:

	Before	After	Delta
Revenue Q2	1,400,000	1,440,000	+2.9%
Revenue Q3	900,000	1,728,000	+92.0%
Revenue Q4	1,035,000	2,073,600	+100.3%
Costs Q1	660,000	600,000	-9.1%
Costs Q2	770,000	720,000	-6.5%
Costs Q3	720,000	864,000	+20.0%
Costs Q4	569,250	1,036,800	+82.1%
Next steps: You can now analyze or further plan using demo-exec-revenue-recovery. Let me know if you want a summary analysis or have follow-up questions!			

> *Run a summary analysis on demo-exec-revenue-recovery to see if the recovery numbers hold up.*

Summary analysis of demo-exec-revenue-recovery (20% growth, 50% cost ratio): Metrics Overview:

- Revenue (USD)

    - Total: $6,441,600
    - Mean: $1,610,400
    - Min: $1,200,000 (Q1 2026)
    - Max: $2,073,600 (Q4 2026)
    - Trend: Up (largest swing: Q1 → Q2 at +20.0%)
- Costs (USD)
    - Total: $3,220,800
    - Mean: $805,200
    - Min: $600,000 (Q1 2026)
    - Max: $1,036,800 (Q4 2026)
    - Trend: Up (largest swing: Q1 → Q2 at +20.0%)

Anomalies (threshold: 50%): None detected. Overall: The recovery model's numbers are consistent, with no anomalies. The plan shows strong upward trends for both Revenue and Costs, and achieves the recovery target for Q4. If you want a deeper dive into any period or metric, just ask!

> *Give strategic recommendations for the recovery scenario demo-exec-revenue-recovery to reach $1.8M in Q4 while improving margins.*

Here are strategic recommendations for the demo-exec-revenue-recovery scenario to reach at least $1.8M revenue in Q4 2026 while improving margins: Model Overview:

- Revenue (USD): total = $6,441,600, mean = $1,610,400, trend: +72.8%, first = $1,200,000, last = $2,073,600
- Costs (USD): total = $3,220,800, mean = $805,200, trend: +72.8%, first = $600,000, last = $1,036,800 Strategic Recommendations:

1. With the goal of reaching at least $1.8M in Q4 revenue while improving margins, consider targeting a new total revenue of $7,729,920 (20% above the current model's total of $6,441,600).
2. To improve margins, evaluate cost reduction strategies in addition to revenue growth, such as renegotiating supplier contracts, optimizing operations, and reducing variable costs.
3. Closely monitor both revenue and cost trends each quarter. If revenue is well ahead of $1.8M in Q4's forecast, ensure cost growth is not keeping pace—aim for a lower cost ratio in Q4 (<50%).
4. Use scenario modeling to analyze the effect of different cost ratios (e.g., 45% vs 50%) and growth rates to further widen the margin.
5. Identify and mitigate risks such as customer churn, sales pipeline volatility, or cost inflation that may threaten margin gains.
6. Regularly review and refine sales and marketing strategies to sustain aggressive growth while managing resource constraints.
7. Set up leading indicator metrics (e.g., monthly pipeline velocity, average deal size, cost per acquisition) for early warning signs of margin pressure. Would you like to simulate and analyze the impact of reducing the cost ratio for Q4 or run margin scenarios for further planning?

> *Give strategic recommendations for demo-exec-hiring to right-size the team for a revenue recovery push without overspending.*

Strategic recommendations for the demo-exec-hiring model could not be automatically generated because the model currently lacks actual Headcount and CostPerHire data. This prevents specific calculations and comparisons to support actionable hiring guidance. However, based on best practices for right-sizing your team to support a major revenue recovery push (such as in your demo-exec-revenue-recovery scenario) without overspending, here are tailored recommendations you can use:

——————————————————————————————————————— Strategic Recommendations for Right-Sizing Hiring (Q4 2026):

1. Align Headcount Growth to Revenue Plan Set Q4 2026 headcount targets based on projected workload increases from your aggressive revenue plan. Avoid "overhiring for growth" before realizing sustained demand.
2. Prioritize Revenue-Generating Roles Focus hiring on roles directly contributing to revenue (e.g., sales, delivery, key customer support), deferring back-office expansion unless absolutely necessary.
3. Benchmark CostPerHire & Hiring Productivity Research industry CostPerHire averages to avoid overspending. Track productivity metrics for recent hires and adjust hiring velocity if ramp-up is slow.
4. Monitor Efficiency Metrics Keep a close eye on Revenue per FTE and Operating Margin. If team size grows faster than revenue, pause on additional hires and re-evaluate.
5. Scenario-Plan for Headcount Needs Model a "Freeze" scenario (minimal headcount), "Steady Growth," and "Accelerated Hiring" for Q4 to understand financial impacts, justifying each incremental hire via expected ROI.
6. Control Risks

- Set hiring approval thresholds—require business case reviews for every growth hire.
- Tie hiring decisions to leading indicators (e.g., sales pipeline, customer onboarding rate) rather than lagging results.

7. Agility in Workforce Planning Consider contract or temporary staff for uncertain or short-term needs, minimizing the risk of long-term fixed costs outpacing actual business growth.
8. Feedback Loop Implement a rapid-cycle feedback loop between hiring, sales growth, and financial performance. Adjust headcount plans quarterly, not just annually.
   ——————————————————————————————————————— To unlock automated, scenario-based recommendations: Please provide Headcount and CostPerHire values for Q1–Q3 2026 in the demo-exec-hiring model, or specify likely figures for Q4 scenarios (e.g. target headcount for each path). Would you like to provide this data, or receive instructions for updating the model so you can access even more targeted and quantified advice?

> *Delete the demo-exec-revenue, demo-exec-revenue-recovery, and demo-exec-hiring models.*

All requested models have been successfully deleted:

- demo-exec-revenue
- demo-exec-revenue-recovery
- demo-exec-hiring If you need to start fresh with new models or would like to see your updated model list, just let me know!

> */quit*

Bye.